

# Bigram Models

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- Condition on previous single word
- Generative process:
  - pick START, pick a word conditioned on previous one, repeat until STOP
- The model can be factored thusly:

$$p(x_1 \dots x_n) = \prod_{i=1}^n p(x_i | x_{i-1})$$

- This is equivalent to assuming the words of the sentence are (homogeneous, 1<sup>st</sup> order) Markov chains, namely that:

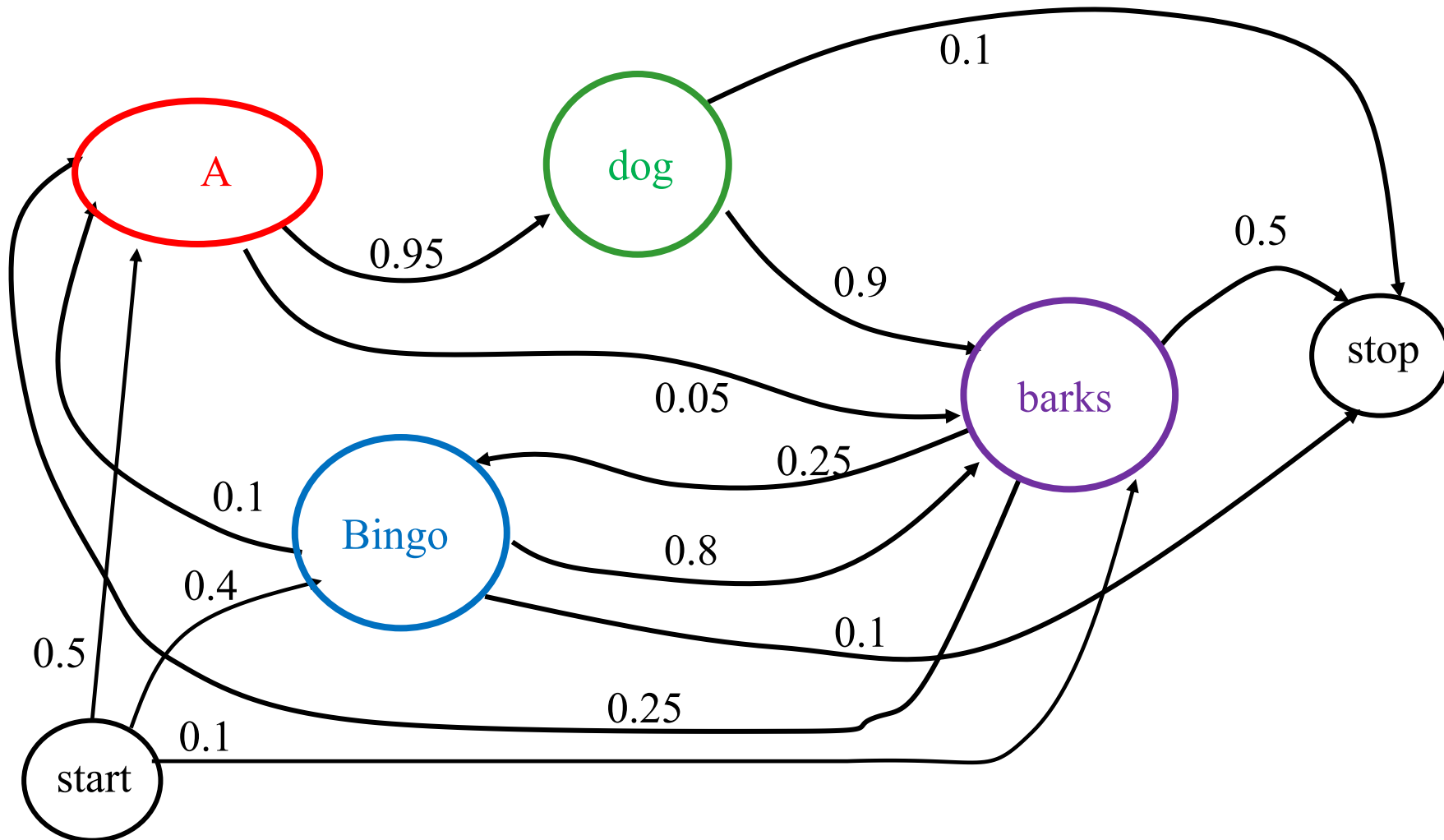
$$p(x_i | x_{i-1} \dots x_0) = p(x_i | x_{i-1})$$

# Markov Model / Markov Chain

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- A finite state automaton with probabilistic state transitions
- Makes Markov assumption that next state only depends on the current state and independent of previous history **given the current state**

# Sample Markov Model for LM



# Formal Representation of Markov LMs

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- The parameters of a Markov LM can be represented as a stochastic, positive-valued matrix (*transition matrix*)
  - Sometimes the distribution of the first symbol is represented separately, but we can think about the sequences as always starting with  $n$  instances of a special START symbol

$$A_{ij} = P(x_m = w_j | x_{m-1} = w_i)$$

$$w_i, w_j \in WORDS \cup \{START, STOP\}$$

- Example:

|        | START | FUNNY | LITTLE | THINGS | STOP |
|--------|-------|-------|--------|--------|------|
| START  | 0     | 0.5   | 0.5    | 0      | 0    |
| FUNNY  | 0     | 0.8   | 0.1    | 0.1    | 0    |
| LITTLE | 0     | 0.1   | 0.1    | 0.8    | 0    |
| THINGS | 0     | 0     | 0      | 0.1    | 0.9  |
| STOP   | 0     | 0     | 0      | 0      | 1    |

# Learning Bigram Language Models

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- A maximum likelihood estimator for the transition matrix is obtained by counting:

$$q_{ML}(x_m = w_j | x_{m-1} = w_i) = \frac{\text{count}(w_i, w_j)}{\sum_{j'} \text{count}(w_i, w_{j'})}$$